PROJECT REPORT

on

Student result system

a report submitted in
partial fulfillment
for the award of

Bachelor's of Computer Applications (Integrated)

By

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1. Introduction

1.1 Introduction to the Project

The Student Result Management System is a console-based application developed in Python to automate the academic record-keeping process. In educational institutions, managing large volumes of student data, calculating marks, and determining results manually can be time-consuming and prone to human error. This project provides a digital solution where student information, such as Roll Numbers, Names, and individual subject marks, can be stored systematically. It utilizes fundamental Data Structures to ensure that data is organized and easily accessible.

1.2 Purpose of the Project

The primary purpose of this project is to simplify the management of student performance data. The specific objectives include:

• Automation: To eliminate manual calculations of total marks and percentages.

• Data Organization: To use nested data structures (lists) to group related student information together.

• Efficiency: To provide a quick search mechanism to retrieve a student's result using their unique Roll Number.

• Skill Application: To demonstrate the practical use of Python programming concepts like loops, conditional branching, and list manipulation as per the BCA 1st-year curriculum.

2. Software Requirement Specifications (SRS)

2.1 Hardware Requirements

Since this is a CLI (Command Line Interface) based Python application, it has very minimal hardware needs. It can run on almost any modern computer:

• Processor: Intel Core i3 or equivalent (Minimum); Intel Core i5 or higher (Recommended).

• RAM: 4 GB (Minimum); 8 GB (Recommended).

• Hard Disk: 100 MB of free space (for Python installation and script storage).

• Input Devices: Standard Keyboard and Mouse.

• Display: Standard Monitor with at least \bm{1024 \times 768} resolution.

2.2 Software Requirements

To run and develop this project, the following software environment is required:

• Operating System: Windows 10/11, macOS, or Linux (Ubuntu/Debian).

• Programming Language: Python 3.10 or higher.

• Development Environment: Visual Studio Code (VS Code), PyCharm, or any standard Text Editor (like Notepad++).

• Version Control: Git (for managing the code) and GitHub (for online hosting and documentation).

3)Display of Outputs(arbitrary inputs):

1. Insertion of data:

```
Enter choice (1-5): 1

--- Enter Student Details ---
Enter Roll Number: 1
Enter Full Name: mayank
Enter Programming Marks: 90
Enter Maths Marks: 80
Enter English Marks: 90

>>> Record added successfully!

================================
  STUDENT RESULT RECORD SYSTEM
================================
1. Add New Student Result
2. Display All Results
3. Search Result by Roll No
4. Delete a Record
5. Exit

Enter choice (1-5): 1

--- Enter Student Details ---
Enter Roll Number: 2
Enter Full Name: aarya
Enter Programming Marks: 8
Enter Maths Marks: 60
Enter English Marks: 100

>>> Record added successfully!
```

2.Display of data:

```
================================
  STUDENT RESULT RECORD SYSTEM
================================
1. Add New Student Result
2. Display All Results
3. Search Result by Roll No
4. Delete a Record
5. Exit

Enter choice (1-5): 2

--- ALL STUDENT RESULTS ---
Roll | Name | Prog | Math | Eng | Total | % | Status
-------------------------------------------------------------
1 | mayank  | 90 | 80 | 90 | 260 | 86.67% | PASS
2 | aarya | 8 | 60 | 100 | 168 | 56.0% | PASS
```

3.Searching of data:

```
Enter choice (1-5): 3

Enter Roll No to search: 2

--- Record Found ---
Name: aarya
Total: 168
Percentage: 56.0%
Status: PASS
```

4.Deletion of data

```
Enter choice (1-5): 1

--- Enter Student Details ---
Enter Roll Number: 4
Enter Full Name: payal
Enter Programming Marks: 50
Enter Maths Marks: 70
Enter English Marks: 80

>>> Record added successfully!

================================
  STUDENT RESULT RECORD SYSTEM
================================
1. Add New Student Result
2. Display All Results
3. Search Result by Roll No
4. Delete a Record
5. Exit

Enter choice (1-5): 4

Enter Roll No to delete: 4

Record for 4 deleted.
```

5.EXITING THE PROGRAM:

```
Enter choice (1-5): 5
Closing the system. Goodbye!
```

4)CONCLUSION:

The Student Result Management System project has been successfully developed and tested. This project was designed to provide a digital solution for schools or colleges to manage student records without the need for manual paperwork. By using Python, I was able to create a system that is both fast and accurate in calculating student performance.

Project Breakdown

During the development of this application, several important programming goals were met:

• Storage Logic: The system uses a master list to hold all student data. Each time a new student is added, their details (Roll Number, Name, and Marks) are grouped into a smaller list and "appended" to the master database. This ensures that the data stays organized and linked together.

• Automated Calculations: One of the biggest advantages of this system is that it removes human error. Once the marks are entered, the program automatically adds them up to find the Total and then calculates the Percentage. It also uses conditional logic to instantly decide if a student has "Passed" or "Failed" based on the 40% rule.

• User Interaction: I designed a simple Menu-Driven Interface. This makes the program very easy to use because the user only has to type a number (1 to 5) to perform any task, whether it is adding a result, viewing the whole list, or searching for a specific record.

• Searching and Deleting: The project includes a search feature that uses a Linear Search method. It looks through the list of roll numbers one by one until it finds a match. Similarly, the delete function allows the user to remove incorrect records easily, which keeps the database clean.

Learning Experience

Working on this project helped me understand how data flows through a program. I learned how to handle user input, how to use loops to keep a program running, and how to use indexing to pull specific information out of a list. It also gave me my first experience using GitHub for hosting my code online, which is a very important skill for any BCA student.

In the future, this project can be improved by adding a login system for security or by saving the data into a permanent file so that it is not deleted when the computer is turned off.

Project link: https://github.com/Mayank-tech226/Student-result-system

ROLE PLAYED BY EVERY MEMBER:

Mayank kumar (0251bca124): Lead Developer & Logic Integration

• Role: Technical Lead

• Responsibilities:

• Wrote the Core Logic of the Python script (The while loop, if-elif branching, and calculation formulas).

• Implemented the Data Structure (Nested Lists) and the Search/Delete Algorithms.

• Ensured data integrity by performing Type Casting (converting inputs to integers).

• Conducted the final Debugging and code optimization to ensure the system runs without errors.

Payal kumari (0251bca121): DevOps & Documentation Specialist

• Role: Deployment Head

• Responsibilities:

- Managed the Version Control by setting up the GitHub Repository.
- Authored the comprehensive README.md file to explain the project professionally.
- Handled the File Management (renaming files, setting up the repository structure, and license).
- Performed Software Requirement Specifications (SRS) research for the report.

Aarya (0251bca131): Quality Assurance & Technical Writing

- Role: Project Coordinator & Reporter
- Responsibilities:
- Compiled the Project Report, including the Introduction, Purpose, and Conclusion.
- Captured and edited Output Screenshots to demonstrate the program's functionality.
- Conducted User Acceptance Testing (UAT) to ensure the menu-driven interface was user-friendly.
- Researched the Hardware Requirements and formatted the final document for submission.

THE END